

Math 520, F15
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score	
1	16		
2	16		
3	16		
4	16		
5	16		
6	20		
Total	100		

Good Luck !

Problem 1. (16 points) Solve the following initial value problem in explicit form and determine the interval in which the solution is defined:

$$y' = \frac{x^2}{y^2 + x^3 y^2}, \quad y(0) = 1.$$

Solution: $\frac{dy}{dx} = \frac{x^2}{(1+x^3)y^2} \Rightarrow y^2 dy = \frac{x^2}{1+x^3} dx$

$$\Rightarrow \int y^2 dy = \int \frac{x^2}{1+x^3} dx \Rightarrow \frac{1}{3} y^3 = \frac{1}{3} \ln|1+x^3| + C$$

$$\Rightarrow y^3 = \ln|1+x^3| + 3C \Rightarrow y = \sqrt[3]{\ln|1+x^3| + 3C}$$

$$y(0) = 1 \Rightarrow 1 = \sqrt[3]{\ln(1+0^3) + 3C} \Rightarrow C = \frac{1}{3}$$

$$\Rightarrow y = \sqrt[3]{\ln|1+x^3| + 1}$$

$1+x^3 \neq 0 \Rightarrow x^3 \neq -1 \Rightarrow x \neq -1$, we get two intervals $(-\infty, -1)$ and $(-1, +\infty)$. Since $x_0 = 0$, we pick up the interval $(-1, +\infty)$ in which the solution is defined.

Problem 2. (16 points) Solve the following initial value problem:

$$t \frac{dy}{dt} = 2y + t^3 \sin t, \quad y(\pi) = 0.$$

Solution: standard form $\frac{dy}{dt} - \frac{2}{t} y = t^2 \sin t$

$$P(t) = -\frac{2}{t}, \quad g(t) = t^2 \sin t$$

Integrating factor: $\mu(t) = e^{\int -\frac{2}{t} dt} = e^{-2 \ln t} = t^{-2} = \frac{1}{t^2}$

thus we have

$$\frac{d}{dt} [\mu(t) y] = g(t) \mu(t).$$

$$y(\pi) = 0 \Rightarrow$$

$$\Rightarrow \left(\frac{1}{t^2}\right) y = \int t^2 \sin t \cdot \left(\frac{1}{t^2}\right) dt \quad 0 = -\pi^2 (\cos \pi + C)$$

$$\Rightarrow C = -\cos \pi = 1$$

$$\frac{y}{t^2} = \int \sin t dt$$

Thus

$$\frac{y}{t^2} = -\cos t + C$$

$$y(t) = -t^2 (\cos t + 1)$$

$$\Rightarrow y = -t^2 (\cos t + 1)$$

Problem 3. (16 points) Given a differential equation

$$(e^{2ax+y} + 3x^2y^2 + x)dx + (2yx^3 + e^{2ax+y})dy = 0$$

where a is a constant.

(a) Determine the constant a such that the above equation is exact.

Solution: $M(x,y) = e^{2ax+y} + 3x^2y^2 + x$, $N(x,y) = 2yx^3 + e^{2ax+y}$

The equation is exact $\Leftrightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$$\frac{\partial M}{\partial y} = e^{2ax+y} + 6x^2y \quad \frac{\partial N}{\partial x} = 6x^2y + (2a)e^{2ax+y}$$

$$\Rightarrow e^{2ax+y} + 6x^2y = 6x^2y + (2a)e^{2ax+y}$$

$$e^{2ax+y} = 2ae^{2ax+y}$$

$$\Rightarrow 1 = 2a \Rightarrow a = \frac{1}{2}$$

(b) Find the general solution of the resulting exact equation in (a).

$$a = \frac{1}{2} \rightarrow (e^{x+y} + 3x^2y^2 + x)dx + (2x^3y + e^{x+y})dy = 0$$

Solution: $M(x,y) = e^{x+y} + 3x^2y^2 + x$, $N(x,y) = 2x^3y + e^{x+y}$

$$\frac{\partial \psi}{\partial x} = M, \quad \frac{\partial \psi}{\partial y} = N$$

$$\begin{aligned} \psi(x,y) &= \int M dx = \int (e^{x+y} + 3x^2y^2 + x) dx \\ &= e^{x+y} + x^3y^2 + \frac{x^2}{2} + h(y) \end{aligned}$$

$$\begin{aligned} \frac{\partial \psi}{\partial y} &= N \Rightarrow e^{x+y} + 2x^3y + h'(y) = 2x^3y + e^{x+y} \\ h'(y) &= y \Rightarrow h(y) = \int y dy = \frac{y^2}{2} \end{aligned}$$

Thus the general solution is given by

$$e^{x+y} + x^3y^2 + \frac{x^2}{2} + \frac{y^2}{2} = C$$

Problem 4. (15 points) Given an initial value problem

$$\frac{dy}{dx} = y - 2x + 1, \quad y(0) = -1.$$

Apply **Euler's method** to approximate the solution of the initial value problem on the interval $[0, 0.4]$ with step size $h = 0.1$.

$$x_0 = 0, h = 0.1, f(x, y) = y - 2x + 1, y_0 = -1, y_{n+1} = y_n + h f(x_n, y_n)$$

$$x_1 = x_0 + h = 0.1$$

$$y_1 = y_0 + h f(x_0, y_0) = -1 + (0.1)[(-1) - 2(0) + 1] = -1$$

$$x_2 = x_1 + h = 0.2$$

$$y_2 = y_1 + h f(x_1, y_1) = -1 + (0.1)[(-1) - 2(0.1) + 1] = -1 - 0.02 = -1.02$$

$$x_3 = x_2 + h = 0.3$$

$$y_3 = y_2 + h f(x_2, y_2) = -1.02 + (0.1)[(-1.02) - 2(0.2) + 1] \\ = -1.02 - 0.042 = -1.062$$

$$x_4 = x_3 + h = 0.4$$

$$y_4 = y_3 + h f(x_3, y_3) = -1.062 + (0.1)[(-1.062) - 2(0.3) + 1] = -1.1282$$

Problem 5. (16 points) Solve the following initial value problem:

$$y'' + 4y' + 13y = 0, \quad y(0) = -4, \quad y'(0) = -1$$

Solution: Characteristic equation: $r^2 + 4r + 13 = 0$

$$r = \frac{-4 \pm \sqrt{4^2 - 4 \cdot 13}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = -2 \pm 3i$$

$$\Rightarrow y(t) = c_1 e^{-2t} \cos 3t + c_2 e^{-2t} \sin 3t$$

$$y'(t) = -2c_1 e^{-2t} \cos 3t - 3c_1 e^{-2t} \sin 3t - 2c_2 e^{-2t} \sin 3t \\ + 3c_2 e^{-2t} \cos 3t$$

$$y(0) = -4 \Rightarrow c_1 = -4$$

$$y'(0) = -1 \Rightarrow -2c_1 + 3c_2 = -1 \Rightarrow 3c_2 = -1 - 8 = -9 \\ c_2 = -3$$

$$\text{Thus } y(t) = -4e^{-2t} \cos 3t - 3e^{-2t} \sin 3t$$

Problem 6. (20 points) Given the following nonhomogeneous equation:

$$y'' + 2y' - 3y = (1 + 2t)e^t.$$

(a) Find the general solution of the corresponding **homogeneous** equation.

Characteristic equation

$$r^2 + 2r - 3 = 0$$

$$(r-1)(r+3) = 0$$

$$\Rightarrow r_1 = 1, r_2 = -3$$

$$\Rightarrow y_c(t) = C_1 e^t + C_2 e^{-3t}$$

(b) Determine a suitable form for the particular solution $Y(t)$ of the given nonhomogeneous equation and then find it by using the **Method of Undetermined Coefficients**.

$$Y(t) = t^s(A+Bt)e^t = Ate^t + Bt^2e^t \quad (s=1)$$

$$Y'(t) = Ae^t + Ate^t + 2Bte^t + Bt^2e^t$$

$$= [A + (A+2B)t + Bt^2]e^t$$

$$Y''(t) = [A+2B+2Bt]e^t + [A+(A+2B)t+Bt^2]e^t$$

$$= [(2A+2B) + (A+4B)t + Bt^2]e^t$$

$$(1+2t)e^t = Y'' + 2Y' - 3Y = [(2A+2B) + (A+4B)t + Bt^2 + 2A + 2(A+2B)t + 2Bt^2 - 3At - 3Bt^2]e^t$$

$$= [(4A+2B) + (8B)t]e^t$$

$$\Rightarrow \begin{cases} 4A+2B=1 \\ 8B=2 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{8} \\ B=\frac{1}{4} \end{cases}$$

$$\Rightarrow Y(t) = \frac{1}{8}te^t + \frac{1}{4}t^2e^t$$

(c) Find the general solution of the nonhomogeneous equation based on results from (a)&(b).

Thus the general solution of the nonhomogeneous equation is

$$y(t) = C_1 e^t + C_2 e^{-3t} + \frac{1}{8}te^t + \frac{1}{4}t^2e^t$$